

Final Project Title

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Abstract

The text of your abstract. 200 or fewer words. It is intended to provide an overview of your paper and may be similar to your synopsis. Note that an abstract can only include one paragraph in this format. You may want to put your synopsis here as you work to refer to it easily.

Keywords: 3 to 6 keywords, that do not appear in the title

*The author gratefully acknowledges *put your thanks here - can wait till final version to input anything, thank friends, peer who did your peer review, parents/guardians, whoever you'd like to acknowledge.*

1 Introduction

Body of paper. Margins in this document are roughly 0.75 inches all around, letter size paper.

This template will be used for you to submit your final project. You need to copy over the PaperTemplate folder in its entirety. Make sure you have the agsm.bst file and bibliography.bib file in order to render to pdf. If you make your own bibliography.bib file later, that's fine - change the name above to match your file name, though for a lot of folks you'll just edit the bibliography.bib file provided. If you don't use a reference in it, it won't print, so you can keep the examples below and drop them at the end and they will disappear (or drop them once you have your own examples in to reference). You'll want the setup for the file to look the same in your repo as in the class repo. You should change the name of the file to something other than "template" at some point though - "draft" or "finalreport", etc when it gets closer to time. Remember that if something goes wrong with the file, you can always look back at the class repo for the original files and their structure, or grab a new copy of the "template" file.

2 Methods

Don't take any of these section titles seriously. They're just for illustration.

3 Verifications

What can we do with the template? References and citations are a big thing to figure out!

We can refer to sections (e.g., this is section [3](#), while the previous section was section [2](#)).

Sub-section references are demo-ed below.

We cite using the BibTeX system. Here we can reference papers by [Campbell & Austin \(2002\)](#) and [Schubert et al. \(2013\)](#) through inline citations (see the `bibliography.bib` file for the reference database).

More work that is relevant can also be cited in a traditional fashion ([Chi et al. 1981](#), [Galyardt \(2014\)](#), [Galyardt \(2012\)](#)).

Note that you can capitalize proper nouns in citations ([Campbell & Austin 2002](#)) (again, see `bibliography.bib`).

We also test some other ways to do the citations, such as [Chi et al. \(1981\)](#), and ([Campbell & Austin 2002](#)). These both work. The latter puts the citation in parentheses. You can also use [Chi et al. \(1981\)](#), though this seems to have similar behavior to just cite.

The original citation examples in the template were:

[Gelman & Vehtari \(2021\)](#) offer some guidance about key ideas about statistical ideas. On an unrelated note, spreadsheets are important to use correctly ([Broman & Woo 2018](#)). Log-linear models are an attractive way to model categorical data ([Bishop et al. 1975](#)).

Note the different structure used in the examples if it is at the end of a sentence/paragraph versus if you are using it as part of the sentence.

Your exposition section (where you have new material) should likely have one citation per paragraph at a minimum where you are pulling from new sources. The idea is to cite where you got your material. Let's look at an example paragraph structure.

This is my intro sentence. I took this idea from source 1. This sentence also paraphrases an idea from source 1 ([Campbell & Austin 2002](#)). Now I am moving onto something else

and it's my own example, so I don't need to cite. Still my own example/words. However, I want to close with something from a different source ([Chi et al. 1981](#)).

If the entire paragraph paraphrases from a single source, it only need a single citation at the end. But you should cite before you go into your own examples, etc. That may mean citing the same source in the same paragraph in different places.

You don't need to cite every sentence, unless swapping sources.

These next sections will be just long enough to illustrate what a full page of text looks like, for margins and spacing.

The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy dog.

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3.1 Example Subsection

You can reference subsections too. This is subsection [3.1](#).

The quick brown fox jumped over the lazy dog. The quick brown fox jumped over the lazy

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4 Examples

Lots of things can be done in this template. Multiple ways of including figures and tables and referencing them are demo-ed below.

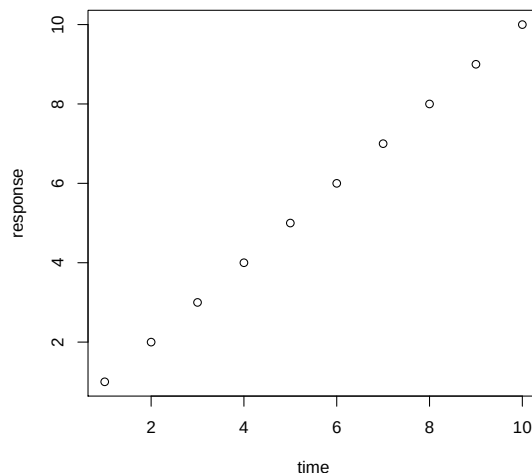


Figure 1: Consistency comparison in fitting surrogate model in the tidal power example.

Table 1: D-optimality values for design X under five different scenarios.

one	two	three	four	five
1.23	3.45	5.00	1.21	3.41
1.23	3.45	5.00	1.21	3.42
1.23	3.45	5.00	1.21	3.43

You can reference these as [Figure 1](#) and [Table 1](#).

What if you are generating the table via code? (Probably what you are doing.) No problem!

```
iris %>%
  group_by(Species) %>%
  summarize(count = n()) %>%
  knitr::kable()
```

Table 2: This is a sample table caption that should interpret the table!

Species	count
setosa	50
versicolor	50
virginica	50

[Table 2](#) displays the number of irises per species. As long as you use the `tbl` and `fig` prefixes, it will keep track and you don't have to!

```
iris %>%
  ggplot(aes(y = Sepal.Width, color = Species)) +
  geom_boxplot()
```

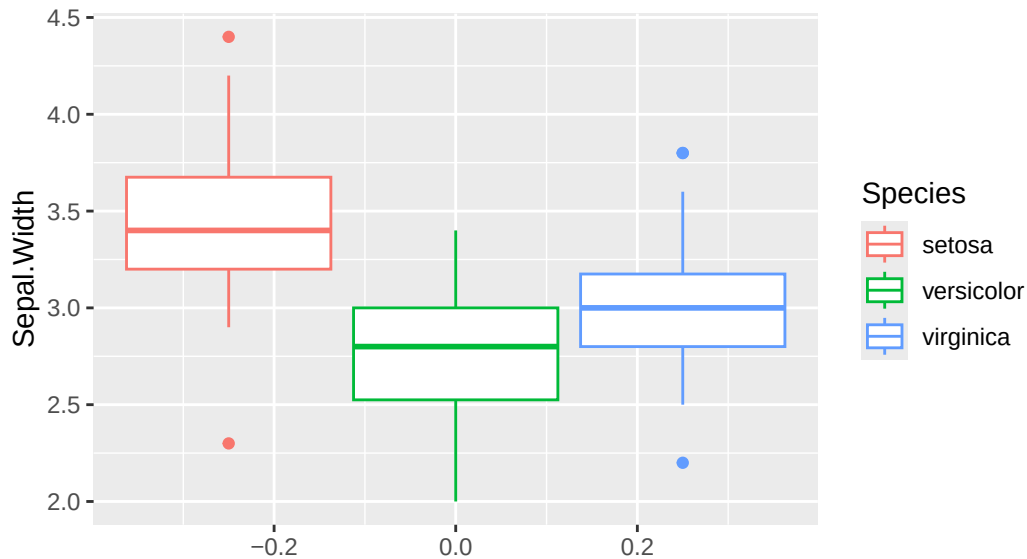


Figure 2: This is a sample figure caption that should interpret the figure!

Similarly, we can reference figures generated by our code. Figure 2 displays the side-by-side boxplots for Sepal.Width by Species.

You can bring in other images if appropriate (need to cite if it's not something you generated). Figure 3 displays a campus map.

```
knitr::include_graphics("campus_map.png")
```

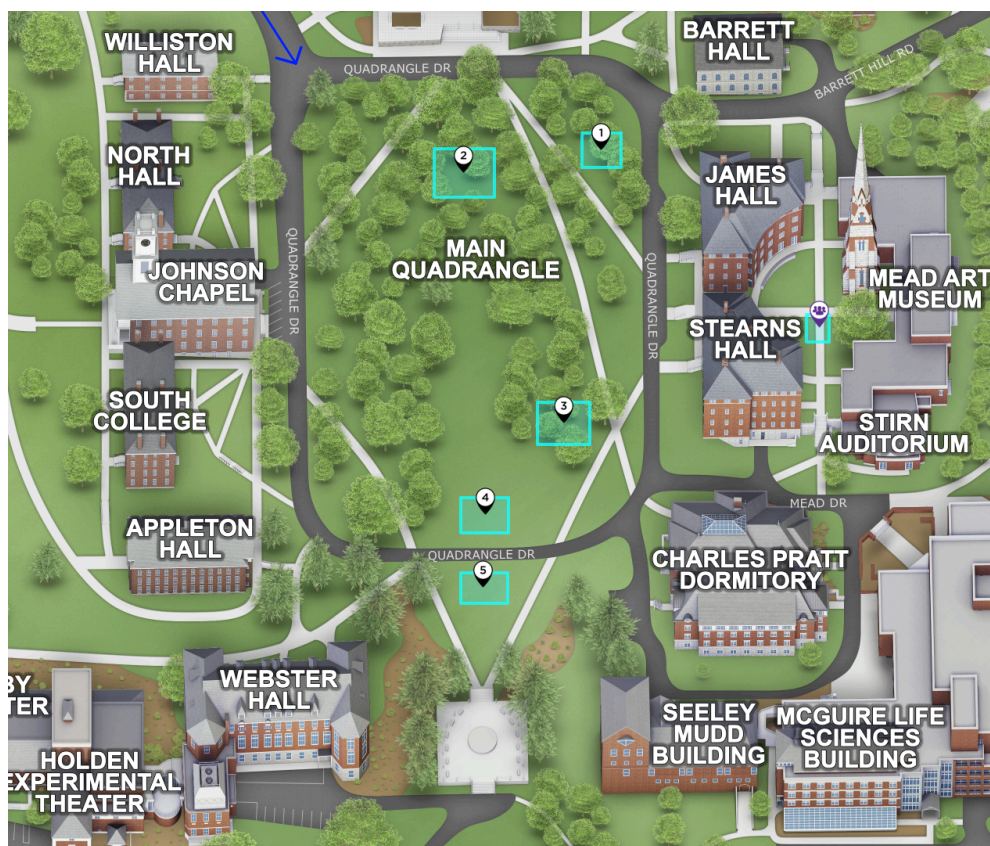


Figure 3: Another sample figure caption

5 LaTeX Examples

To shift into “tex” mode while in-line, it’s pretty easy. You just need dollar signs.

The variables X_1 and X_2 are uncorrelated.

If you have multiple indices or items that need to go into a subscript or a superscript, you need curly brackets.

We want to look at X_{10} and X_{10}^{20} .

If you have a big equation, you probably want to set it on it’s own, rather than try it in-line. That means DOUBLE dollar signs in Qmnds. It doesn’t like a different framework

you might have seen in `.Rmds`. And yes, it seems to “add space”, but that’s ok. Just run with it (if I can figure out how to edit it, will let you know!).

In regression, we assert (for two predictor variables) that

$$\mu_Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2.$$

Don’t forget punctuation in your equations.

If you want to reference equations, you need to be a bit more formal:

$$\mu_Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2. \tag{1}$$

Now I can refer to the equation as Equation 1.

I’m going to borrow a lot of the examples below from the revision of my Probability textbook, so if the context is a little strange (meaning, not from our class), that’s why.

You can do bulleted lists using LaTeX instead of RMarkdown formatting like this:

1. “Define the terms: continuous RV, probability density function, and cumulative density function.
2. Compute expectation and variance for continuous RVs.
3. Apply uniform and exponential distributions to appropriate problems.
4. Solve problems involving joint and marginal distributions in the continuous setting.
5. Extend independence, covariance, and correlation concepts to the continuous setting.

6. Work with uniform and exponential distributions in R to find probabilities and simulate problems.” (Chapter 6 introduction)

In equations, you can also do alignments and adjust spacing with various commands. In text, you can also get italics and bold text. Here is some text with examples of these, including fractions in the equations.

“That is, we integrate a constant c over the bullseye region. This leads to

$$P(B) = \iint_B c \, dx \, dy = c [\text{Area}(B)] = \frac{c\pi}{16}. \quad (2)$$

What is c ? As this is a probability model, the points in the sample space should add up or integrate to 1. The sample space is C , the target. This gives

$$1 = P(C) = \iint_C c \, dx \, dy = c [\text{Area}(C)] = c\pi$$

and thus $c = 1/\pi$. Plugging in c to Equation 2 gives

$$P(B) = \frac{1}{\text{Area}(C)} \iint_B dx \, dy = \frac{1}{16},$$

the proportion of the total area of the target taken up by the bullseye.

This gives the beginnings of a *continuous uniform probability model*. If Ω is a continuous set with all points equally likely, then for subsets $S \subseteq \Omega$,

$$P(S) = \frac{1}{\text{Area}(\Omega)} \iint_S dx \, dy = \frac{\text{Area}(S)}{\text{Area}(\Omega)}.$$

In one dimension, the double integral becomes a single integral and area becomes length.

In three dimensions we have a triple integral and volume. (Chapter 6 introduction)''

Here is an example from Chapter 6 with an alignment actually needed for the two lines (the first equals sign in each line lines up):

$$\begin{aligned} 1 &= \int_{-\infty}^{\infty} ce^{-|x|} dx = \int_{-\infty}^0 ce^x dx + \int_0^{\infty} ce^{-x} dx \\ &= 2c \int_0^{\infty} e^{-x} dx = 2c (-e^{-x})|_0^{\infty} = 2c, \end{aligned}$$

Here is an example from Chapter 6 with an array (for alignment) and text inserted into the equation using mbox.

A random variable X has density function

$$f(x) = \begin{cases} 2/5, & \text{if } 0 < x \leq 1 \\ 2x/5, & \text{if } 1 \leq x < 2 \\ 0, & \text{otherwise.} \end{cases}$$

I think most of you will use kable or gtsummary for generating tables, but if you want to make tables in LaTeX, you can do that too. Ask me for assistance if you want to do that - I'm not including examples here other than what the template came with. Tables are much easier to do in R using existing packages, imo, so I would recommend that over LaTeX anyway.

6 Conclusion

You are responsible for choosing the structure you want for the report, i.e. what sections and their names. Here are some examples that you could adapt, as appropriate. These are just examples - the first has more sections, some of which may make more sense as sub-sections of others. I'm just trying to give you examples. You don't have to have a conclusion or an introduction, etc. You'll need to find what works for you. Here are some examples. Again - you get to determine how many sections you have and what they are called!

6.1 Example Structure 1 - Many sections

- Introduction
- Background
- Topic X (Main exposition section)
- Applications in Literature
- Toy Example
- Application to Data (your data)
- Conclusion

6.2 Example Structure 2 - Few sections, more subsections

- Introduction (including any relevant Background)
- Topic X (exposition, examples in literature, and toy example of R package fns)
- Simulation Study

References

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